

Solving Recurrence Relations using Substitution Method

Introduction

This handout presents step-by-step examples of solving recurrence relations using the **substitution method**. The substitution (or induction) method involves guessing a solution and then proving it by mathematical induction.

Example 1: $T(n) = 2T(n/2) + n$

Guess: $T(n) = O(n \log n)$

Inductive Hypothesis: Assume $T(k) \leq ck \log k$ for all $k < n$

Proof:

$$\begin{aligned} T(n) &= 2T(n/2) + n \\ &\leq 2(c(n/2) \log(n/2)) + n \\ &= cn \log(n/2) + n = cn(\log n - 1) + n \\ &= cn \log n - cn + n \leq cn \log n \quad \text{if } c \geq 1 \end{aligned}$$

Therefore, $\boxed{T(n) = O(n \log n)}$

Example 2: $T(n) = T(n/2) + \log n$

Guess: $T(n) = O(\log^2 n)$

Inductive Hypothesis: $T(k) \leq c \log^2 k$

Proof:

$$\begin{aligned} T(n) &= T(n/2) + \log n \\ &\leq c \log^2(n/2) + \log n \\ &= c(\log n - 1)^2 + \log n \\ &= c(\log^2 n - 2 \log n + 1) + \log n \\ &= c \log^2 n - 2c \log n + c + \log n \\ &= c \log^2 n - (2c - 1) \log n + c \end{aligned}$$

Choose c large enough (e.g., $c \geq 2$) so the negative term doesn't dominate.

Therefore, $\boxed{T(n) = O(\log^2 n)}$

Example 3: $T(n) = 3T(n/4) + n$ **Guess:** $T(n) = O(n)$ **Inductive Hypothesis:** $T(k) \leq ck$ **Proof:**

$$\begin{aligned}
T(n) &= 3T(n/4) + n \\
&\leq 3c(n/4) + n = \frac{3c}{4}n + n = n \left(\frac{3c}{4} + 1 \right)
\end{aligned}$$

We want $\frac{3c}{4} + 1 \leq c \Rightarrow 1 \leq c - \frac{3c}{4} = \frac{c}{4} \Rightarrow c \geq 4$

Therefore, $\boxed{T(n) = O(n)}$ **Example 4:** $T(n) = T(n-1) + \log n$ **Guess:** $T(n) = O(n \log n)$ **Inductive Hypothesis:** $T(k) \leq ck \log k$ **Proof:**

$$\begin{aligned}
T(n) &= T(n-1) + \log n \\
&\leq c(n-1) \log(n-1) + \log n \\
&\leq c(n-1) \log n + \log n = (c(n-1) + 1) \log n \\
&= (cn - c + 1) \log n \leq cn \log n \quad \text{if } c \geq 1
\end{aligned}$$

Therefore, $\boxed{T(n) = O(n \log n)}$ **Example 5:** $T(n) = T(n/2) + T(n/4) + T(n/8) + n$ **Guess:** $T(n) = O(n)$ **Inductive Hypothesis:** $T(k) \leq ck$ **Proof:**

$$\begin{aligned}
T(n) &= T(n/2) + T(n/4) + T(n/8) + n \\
&\leq c(n/2) + c(n/4) + c(n/8) + n \\
&= c \left(\frac{7n}{8} \right) + n = n \left(\frac{7c}{8} + 1 \right)
\end{aligned}$$

We want $\frac{7c}{8} + 1 \leq c \Rightarrow 1 \leq c - \frac{7c}{8} = \frac{c}{8} \Rightarrow c \geq 8$

Therefore, $\boxed{T(n) = O(n)}$